ITAMAR AMSALEM

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EDUCATION

Rutgers University — **New Brunswick, New Jersey**
B.S. in Mathematics, B.A. in Computer Science — *September 2024 – Dec 2027*

- **GPA:** 3.61 / 4.00 | Dean's List All Semesters
- **Selected Coursework:** Intro to Mathematical Finance, Intro to Stochastic Processes, Machine Learning Principles, Mathematical Theory of Probability, Statistics II, Linear Optimization, Linear Algebra, Multivariable Calculus

PROFESSIONAL EXPERIENCE

AQC Tech — **Lakewood, New Jersey**
Quantitative Research Intern — *June 2026 – August 2026*

- Built an ML pipeline forecasting BTC execution prices from live order-book data to power a large-order RFQ pricing strategy, reconstructing 100+ days of raw order-book history into a training-ready dataset from scratch
- Engineered time-series features (rolling momentum, drawdown/drawup, z-score) and found a change to the underlying signal that increased the model's win rate against a strict baseline from ~1-3 to 7 of 25 held-out days; built a leak-proof evaluation framework (leave-one-day-out cross-validation, baseline stress-testing) to catch misleadingly optimistic results before they shipped
- Compared gradient-boosted trees, random forest, and neural network models for the pricing task, analyzing each model's training dynamics, hyperparameter sensitivity, and computational tradeoffs (GBM's sequential residual-fitting vs. RF's parallelizable bootstrap ensembling), then ran hyperparameter and sensitivity sweeps to pick and stress-test the best-performing model

PROJECTS

SurfaceSense — **Rutgers Road to Silicon Valley — 2nd Place**
Fleet-Mounted LiDAR Street-Data Startup — *October 2026*

- Co-founded a street-data startup that mounts LiDAR and camera kits on municipal garbage trucks and street sweepers, measuring sidewalk and road hazards in 3D and tracking changes weekly
- Designed a two-sided revenue model: SaaS dashboards for towns priced against existing survey spend, plus data licenses for utilities, insurers, and mapping companies, with a path to $18M in annual revenue
- Pitched a $500K pre-seed to fund 10 New Jersey pilots anchored by Princeton, targeting ADA-mandated sidewalk spending ($1.55B in NYC, $1.37B in LA)

Trade the Knight — **Rutgers QFC Algorithmic Trading Competition — 1st Place**
Algorithmic Trading Competition — *April 2026*

- Engineered and deployed two concurrent statistical arbitrage algorithms via pairs trading, executing live against a proprietary Rust-based exchange with a custom matching engine, realistic latency, and live order books
- Achieved ~100% return on $1M notional with a 2.26 Sharpe ratio over 8 hours of live trading; alpha generated through systematic position sizing, real-time risk controls, and multi-strategy coordination
- Placed 1st out of 80 competing traders; competition sponsored by Citadel, Optiver, and ThetaData

GSD.AI — **IBM Datathon — 1st Place**
AI-Powered Restaurant Survival & Market Intelligence Platform — *February 2026*

- Trained an XGBoost classifier to predict whether a new restaurant concept will survive 5+ years based on 40+ engineered features (location, cuisine type, local competition); achieved 91.9% accuracy and 0.82 ROC-AUC
- Scored every NJ zip code on how well it fits a given restaurant concept using ML-derived demand gap and market size signals from 163K+ Yelp reviews across 91 zip codes
- Deployed an AI-powered recommendation web app where restaurant owners enter their concept and receive ranked location matches across all of NJ; built the entire platform end-to-end in 24 hours, placing 1st

TECHNICAL SKILLS

- *Quantitative Skills:* AI/ML Pipelines, Regression Analysis, Classification Models, Time-Series Analysis, Feature Engineering, Backtesting, Cross-Validation, Data Wrangling, Risk Modeling
- *Languages:* C, Python (pandas, NumPy, scikit-learn, XGBoost, SHAP, LLM APIs), Java
- *Tools*: Excel, Jupyter Notebook, Git, Flask, FastAPI, Docker
- *Other:* Hebrew (fluent)

ADDITIONAL INFORMATION

Campus Involvement: Rutgers Quant Finance Club, Founding Father & Academic Chair at Alpha Epsilon Pi
Interests: Maccabi Tel Aviv, Formula 1, Poker, Chess, Alpine Skiing, Manchester United